

Simultaneous inference in the context of one-way ANOVA

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LSM

Find confidence intervals with family wise confidence level $(1-\eta)$ for parameters $\theta_1, \dots, \theta_g$ where each θ_j is a contrast involving the treatment means (i.e., each parameter is of the

form $\sum_{i=1}^r c_i \mu_i$ with $c_i = 0$)

$$P(\theta_j \in C_{j,g}^{1-\eta}(\underline{Y}) \text{ for } j=1, \dots, g) \geq 1-\eta$$

Where each $C_{j,g}^{1-\eta}(\underline{Y})$ is an interval depending on $\underline{Y}$

Bonferroni's procedure Take $C_{j,g}^{1-\eta}(\underline{Y})$ to be the set

$$\hat{\theta}_j \pm t_{N-r}^{1-\eta/2g} s(\hat{\theta}_j) \quad \text{where } N = \sum_{i=1}^r n_i$$

Downside of Bonferroni's procedure is that the intervals become larger with increasing g .

Improved strategy due to Tukey for dealing with pairwise comparison (Tukey procedure for simultaneous inference)

Parameters

$$D_{ij} = \mu_i - \mu_j \text{ for } 1 \leq i < j \leq r$$

Suppose further that the design is balanced i.e. $n_1 = \dots = n_r = m$

We try to construct confidence intervals of the form

$$\hat{D}_{ij} \pm T^{1-\eta} s(\hat{D}_{ij})$$

The statement that $D_{ij} \in C_{ij,T}^{1-\alpha}(\underline{Y})$ for all $1 \leq i < j \leq r$ means

$$\frac{|\hat{D}_{ij} - D_{ij}|}{s(\hat{D}_{ij})} \leq T^{1-\alpha}$$

$$\begin{aligned} \hat{D}_{ij} - D_{ij} &= (\bar{Y}_{i\cdot} - \bar{Y}_{j\cdot}) - (\mu_i - \mu_j) \\ &= (\bar{Y}_{i\cdot} - \mu_i) - (\bar{Y}_{j\cdot} - \mu_j) = W_i - W_j \\ &\sim N(0, \frac{\sigma^2}{n_i}) \end{aligned}$$

$$|W_i - W_j| \leq \max_{1 \leq i < j \leq r} |W_i - W_j|$$

$$= \max_{1 \leq i \leq r} W_i - \min_{1 \leq i \leq r} W_i$$

Under a balanced design $W_i \overset{iid}{\sim} N(0, \frac{\sigma^2}{m})$

$$\text{In general } s^2(\hat{D}_{ij}) = \underset{\substack{\uparrow \\ \text{MSE}}}{\hat{\sigma}^2} \left(\frac{1}{n_i} + \frac{1}{n_j} \right) = \frac{2\hat{\sigma}^2}{m}$$

Summary: Under a balanced design

$$\max_{1 \leq i < j \leq r} \frac{|\hat{D}_{ij} - D_{ij}|}{s(\hat{D}_{ij})} \leq \frac{\max_{1 \leq i \leq r} \left(\frac{W_i}{\sigma} \right) - \min_{1 \leq i \leq r} \left(\frac{W_i}{\sigma} \right)}{\sqrt{\frac{\hat{\sigma}^2}{\sigma^2}} \sqrt{\frac{2}{m}}}$$

$$\cup = \frac{1}{\sqrt{2}} \frac{\max_{1 \leq i \leq r} \sqrt{\frac{m}{\sigma^2}} W_i - \min_{1 \leq i \leq r} \sqrt{\frac{m}{\sigma^2}} W_i}{\sqrt{\frac{SSE}{\sigma^2} \frac{1}{N-r}}} \quad \left. \begin{array}{l} N=r \\ \text{indep} \end{array} \right\}$$

Numerator = range of $Z_1, \dots, Z_r$ where $Z_i \stackrel{iid}{\sim} N(0,1)$

Denominator $\sim \sqrt{\chi^2_{N-r}/N-r}$

If $Z_1, \dots, Z_r \stackrel{iid}{\sim} N(0,1) \rightarrow \text{indep}$
 $V \sim \chi^2_r$

then $\frac{\max_i Z_i - \min_i Z_i}{\sqrt{V/r}}$ is called a studentized range distribution (or Tukey distribution) with degrees of freedom (r, r)

So under a balanced design, if we set $T^{1-\eta}$ to be

$\left(\frac{1}{\sqrt{2}} q^{1-\eta}_{r, N-r} \right)$ then we get simultaneous C.I. for all D_{ij} at familywise confidence level $(1-\eta)$
 $\rightarrow (1-\eta)$ quantile of Tukey $(r, N-r)$ distn.
 $\rightarrow$ multiplier associated with Tukey's procedure

Remark 1: We can use Tukey's multiplier even when we consider smaller than $\binom{r}{2}$ pairwise comparisons

Remark 2: Even if the design is unbalanced, Tukey's procedure, i.e. by setting confidence interval for $D_{ij} = \mu_i - \mu_j$ as $\hat{D}_{ij} \pm T^{1-\eta} s(\hat{D}_{ij})$ for $1 \leq i < j \leq r$

where $T^{1-\eta} = \frac{1}{\sqrt{2}} q_{r, N-r}^{1-\eta}$

and $s(\hat{D}_q) = \sqrt{MSE} \sqrt{\frac{1}{n_i} + \frac{1}{n_j}}$

we can still ensure $(1-\eta)$ familywise confidence level.

Simulate $\begin{pmatrix} X_1 \\ \vdots \\ X_r \end{pmatrix}$ s.t. $X_i \stackrel{\text{indep}}{\sim} N(0, \frac{1}{n_i})$

Constructing simultaneous confidence sets for arbitrary number of contrasts (L)

Let $\mathcal{L} = \left\{ \sum_1^r c_i \mu_i = 0 : \sum_1^r c_i = 0 \right\}$ denote the set of all contrasts

We want the individual confidence intervals to be of the form $\hat{L} \pm S^{1-\eta} s(\hat{L})$

where $(1-\eta)$ = familywise confidence level

$$\hat{L} = \sum c_i \hat{\mu}_i = \sum c_i \bar{Y}_i$$

$$s(\hat{L}) = \sqrt{MSE} \sqrt{\sum \frac{c_i^2}{n_i}}$$

Scheffe's procedure Set $S^{1-\eta} = \sqrt{(r-1) F_{r-1, N-r}^{1-\eta}}$

$(1-\eta)$ quantile of $F_{r-1, N-r}$

What is $\hat{L} - L$

$$= \sum_1^r c_i \bar{Y}_i - \sum_1^r c_i \mu_i$$

$$Y_{ij} = \mu_i + \varepsilon_{ij}$$

$$\bar{\varepsilon}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} \varepsilon_{ij}$$

$$= \sum_1^r c_i (\bar{Y}_i - \mu_i) = \sum_1^r c_i \bar{\varepsilon}_i = \sum_1^r c_i (\bar{\varepsilon}_i - \bar{\varepsilon}_{..})$$

since $\sum c_i = 0$

Thus $\frac{|\hat{L} - L|}{S(\hat{L})} = \frac{|\sum c_i (\bar{\epsilon}_{i\cdot} - \bar{\epsilon}_{\cdot\cdot})|}{S(\hat{L})} \leq \frac{\sqrt{\sum \frac{c_i^2}{n_i}} \sqrt{\sum n_i (\bar{\epsilon}_{i\cdot} - \bar{\epsilon}_{\cdot\cdot})^2}}{\sqrt{\frac{SSE}{N-r}} \sqrt{\sum \frac{c_i^2}{n_i}}}$
 $SST_0 = SST_R + SSE$
 $\quad \quad \quad \swarrow \quad \searrow$
 $\quad \quad \quad \text{indep}$
 (Cauchy-Schwartz)

$$SSTR = \sum n_i (\bar{Y}_{i\cdot} - \bar{Y}_{\cdot\cdot})^2 \sim \sigma^2 \chi_{r-1}^2 \text{ under } H_0: \mu_1 = \dots = \mu_r$$

$$\text{So, } \sum n_i (\bar{\epsilon}_{i\cdot} - \bar{\epsilon}_{\cdot\cdot})^2 \sim \sigma^2 \chi_{r-1}^2$$

Summary

$$\max_{L \in \mathcal{L}} \frac{|\hat{L} - L|}{S(\hat{L})} = \sqrt{\frac{\sum n_i (\bar{\epsilon}_{i\cdot} - \bar{\epsilon}_{\cdot\cdot})^2}{SSE/(n-r)}} \stackrel{\Delta}{=} \sqrt{\frac{\sigma^2 \chi_{r-1}^2}{\sigma^2 \chi_{N-r/N-r}^2}}$$

↑
set of all contrasts

($\because c_i = K n_i (\bar{\epsilon}_{i\cdot} - \bar{\epsilon}_{\cdot\cdot})$,
equality case of C-S)

$$\stackrel{\Delta}{=} \sqrt{(r-1) F_{r-1, N-r}}$$